SYMBOL SPACES FOR BOUNDED AND SCHATTEN COMMUTATORS OF THE DOWNWARD HAAR SHIFT

1. Notation and symbol spaces

We work on $L^2(\mu)$ with the standard dyadic grid $\mathcal{D}$ on $\mathbb{R}$. For $I \in \mathcal{D}$, I_- and I_+ denote its left and right children, $\widehat{I}$ its parent, and $\ell(I)$ its length. We write $\mathbf{1}_A$ for the indicator of A and $\#F$ for the cardinality of a finite set F. The notation $\mathrm{Meas}_+(\mathbb{R})$ means the positive Borel measures on $\mathbb{R}$.

$$\mathcal{D}_k = \{[j2^{-k}, (j+1)2^{-k}) : j \in \mathbb{Z}\}, \qquad \mathcal{D} = \bigcup_{k\in\mathbb{Z}} \mathcal{D}_k, \qquad \mathrm{ch}(I) = \{I_-, I_+\},$$
$$\mathfrak{M} = \{\mu \in \mathrm{Meas}_+(\mathbb{R}) : 0 < \mu(I) < \infty \ (I \in \mathcal{D})\},$$
$$\mathfrak{M}_\infty = \{\mu \in \mathfrak{M} : \mu(\mathbb{R}) = \infty\} \subset \mathfrak{M}, \qquad \mu \in \mathfrak{M}.$$

All results below hold for $\mu \in \mathfrak{M}$ and hence also with the additional standing assumption $\mu(\mathbb{R}) = \infty$. The measure-adapted Haar functions and the downward Haar shift $H = H_{\mathcal{D}}$ are given by

$$m(I) = \frac{\mu(I_-)\mu(I_+)}{\mu(I)}, \qquad h_I = \sqrt{m(I)}\left(\frac{\mathbf{1}_{I_-}}{\mu(I_-)} - \frac{\mathbf{1}_{I_+}}{\mu(I_+)}\right),$$
$$\langle h_I, h_J\rangle = \delta_{I,J}, \qquad \sigma_{I_-} = 1, \quad \sigma_{I_+} = -1, \qquad Hh_I = h_{I_-} - h_{I_+}, \quad H^*h_I = \sigma_I h_{\widehat{I}}.$$

The inner product is linear in the first variable. The notation T^* denotes the Hilbert-space adjoint and $T^{\mathfrak{t}}$ the matrix transpose in the specified orthonormal basis; $\operatorname{rank} T$ is the dimension of its range. In particular,

$$\langle f, g\rangle = \int f\overline{g}\, d\mu, \qquad Hf = \sum_{I\in\mathcal{D}} \langle f, h_I\rangle (h_{I_-} - h_{I_+}), \qquad f \in L^2(\mu).$$

The Haar family is completed by constants on dyadic components of finite measure. We use the following notation for the resulting orthogonal decomposition and complete basis:

$$E_- = (-\infty, 0), \qquad E_+ = [0, \infty), \qquad \mathscr{E} = \{E \in \{E_-, E_+\} : \mu(E) < \infty\},$$
$$\mathcal{H}_h = \overline{\mathrm{span}}\{h_I : I \in \mathcal{D}\}, \qquad e_E = \frac{\mathbf{1}_E}{\sqrt{\mu(E)}} \quad (E \in \mathscr{E}),$$
$$\mathcal{H}_0 = \mathrm{span}\{e_E : E \in \mathscr{E}\} = \mathcal{H}_h^\perp, \qquad L^2(\mu) = \mathcal{H}_h \oplus \mathcal{H}_0,$$
$$H|_{\mathcal{H}_0} = H^*|_{\mathcal{H}_0} = 0, \qquad H^*H = 2P_h, \qquad \|H\| = \sqrt{2}, \qquad P_0 = 1 - P_h,$$
$$\mathcal{I} = \mathcal{D} \sqcup \mathscr{E}, \qquad e_I = h_I \ (I \in \mathcal{D}), \qquad \mathscr{F} = \mathrm{span}\{e_u : u \in \mathcal{I}\},$$
$$F \Subset A \iff F \subset A,\ \#F < \infty, \qquad P_A f = \sum_{u\in A} \langle f, e_u\rangle e_u \quad (A \subseteq \mathcal{I}), \qquad \overline{\mathscr{F}} = L^2(\mu).$$

Here P_h and P_0 are the orthogonal projections onto $\mathcal{H}_h$ and $\mathcal{H}_0$, respectively. For any set of basis indices A, P_A is the corresponding orthogonal projection. The notation $F \Subset A$ means that F is a finite subset of A; $\mathscr{F}$ is the space of finite linear combinations of basis vectors. Matrices act on $\mathscr{F}$ by the finite sesquilinear pairing

$$T = [t_{u,v}] \in \mathbb{C}^{\mathcal{I}\times\mathcal{I}}, \quad f, g \in \mathscr{F} : \qquad \langle Tf, g\rangle := \sum_{u,v\in\mathcal{I}} t_{u,v}\langle f, e_v\rangle\overline{\langle g, e_u\rangle}.$$

Date: 2026-11-04.

Thus $t_{u,v} = \langle Te_v, e_u\rangle$ whenever T represents an operator. We write $\mathcal{B}(\mathcal{H})$ and $\mathcal{K}(\mathcal{H})$ for the bounded and compact operators on $\mathcal{H} = L^2(\mu)$. For $T \in \mathcal{K}(\mathcal{H})$, $s_n(T)$ are the eigenvalues of $|T|$, listed in nonincreasing order with multiplicity. These are the singular values of T. For $0 < p < \infty$,

$$|T| = (T^*T)^{1/2}, \qquad S_p(\mathcal{H}) = \left\{ T \in \mathcal{K}(\mathcal{H}) : \sum_{n\ge 1} s_n(T)^p < \infty \right\}, \qquad \|T\|_{S_p} = \left(\sum_{n\ge 1} s_n(T)^p \right)^{1/p}.$$

For $0 < p < 1$ the expression $\|\cdot\|_{S_p}$ is a quasi-norm; for $p \ge 1$ it is a norm. The same notation is used for finite matrices. An unsubscripted operator norm means $\|\cdot\|_{L^2(\mu)\to L^2(\mu)}$. We use $A \lesssim_p B$ and $A \asymp_p B$ for comparison constants depending only on p; without a subscript the constants are absolute.

For a locally integrable complex symbol φ, f_I denotes a local average and $\mathbb{E}_k$ the dyadic conditional expectation. The coefficients b_I and d_I are, respectively, its Haar coefficients and the diagonal entries of its multiplication matrix:

$$\begin{gathered} \varphi \in L^1_{\mathrm{loc}}(\mu;\mathbb{C}), \qquad f_I = \frac{1}{\mu(I)} \int_I f \, d\mu, \qquad \mathbb{E}_k f = \sum_{I\in\mathcal{D}_k} f_I \mathbf{1}_I, \\ b_I = \int \varphi h_I \, d\mu, \\ d_I = \int \varphi h_I^2 \, d\mu = \frac{\mu(I_+)\varphi_{I_-} + \mu(I_-)\varphi_{I_+}}{\mu(I)}, \\ d_I - \varphi_I = \frac{\mu(I_+) - \mu(I_-)}{\mu(I)} (\varphi_{I_-} - \varphi_{I_+}). \end{gathered} \tag{1}$$

In particular, d_I is a reverse-weighted average of the two child means. We write $M_\varphi f = \varphi f$ on its natural domain $\{f \in L^2(\mu) : \varphi f \in L^2(\mu)\}$. For a general locally integrable symbol the commutator is initially defined by its form on $\mathscr{F}$. Membership in an operator class means that this form has an extension in that class:

$$\begin{gathered} c_\varphi(f,g) = \int \varphi\big((Hf)\overline{g} - f\,\overline{H^*g}\big)\, d\mu, \qquad f,g \in \mathscr{F}, \\ C_\varphi \in \mathcal{B}(L^2(\mu)) \iff \exists C \in \mathcal{B}(L^2(\mu)) \ \forall f,g \in \mathscr{F} : \ \langle Cf, g\rangle = c_\varphi(f,g), \\ C_\varphi \in S_p \iff \exists C \in S_p \ \forall f,g \in \mathscr{F} : \ \langle Cf, g\rangle = c_\varphi(f,g), \\ \varphi \in L^\infty(\mu) \implies C_\varphi = [M_\varphi, H] = M_\varphi H - H M_\varphi. \end{gathered} \tag{2}$$

For each finite $F \subset \mathcal{D}$, $G_F(\varphi)$ records the correlations of the averaging vectors u_I. The trace is unnormalized, and powers of a positive matrix are defined by spectral calculus.

$$\begin{gathered} u_I = \frac{\mathbf{1}_I}{\mu(I)}, \qquad G_F(\varphi) = \left[b_I \overline{b_J} \frac{\mu(I\cap J)}{\mu(I)\mu(J)} \right]_{I,J\in F}, \quad F \Subset \mathcal{D}, \\ z^* G_F(\varphi) z = \int \left| \sum_{I\in F} \overline{b_I} z_I u_I \right|^2 d\mu \ge 0, \qquad \operatorname{Tr} A = \sum_{I\in F} A_{I,I}. \end{gathered} \tag{3}$$

We define the two symbol quantities and their associated homogeneous space by

$$\mathfrak{C}_p(\varphi) = \sup_{F\Subset\mathcal{D}} \left[\operatorname{Tr}\left(G_F(\varphi)^{p/2} \right) \right]^{1/p}, \quad 0 < p < \infty, \tag{C}$$

$$\mathfrak{D}_p(\varphi) = \left[\sum_{I\in\mathcal{D}} \left(|d_{I_-} - d_I|^2 + |d_{I_+} - d_I|^2 \right)^{p/2} \right]^{1/p}. \tag{D}$$

$$\mathcal{X}_p(\mu) = \{\varphi \in L^1_{\mathrm{loc}}(\mu) : \mathfrak{C}_p(\varphi) + \mathfrak{D}_p(\varphi) < \infty\}, \qquad \|\varphi\|_{\mathcal{X}_p} = \mathfrak{C}_p(\varphi) + \mathfrak{D}_p(\varphi),$$
$$\left\|\varphi + c_- \mathbf{1}_{E_-} + c_+ \mathbf{1}_{E_+}\right\|_{\mathcal{X}_p} = \|\varphi\|_{\mathcal{X}_p}, \qquad c_-, c_+ \in \mathbb{C}.$$

The last identity expresses invariance under addition of a constant on either dyadic component. The quantity $\|\cdot\|_{\mathcal{X}_p}$ is a seminorm for $p \geq 1$ and a quasi-seminorm for $0 < p < 1$.

2. Schatten characterization for $1 < p < \infty$

Theorem 1. *Let $\mu \in \mathfrak{M}$, $1 < p < \infty$, and $\varphi \in L^1_{\mathrm{loc}}(\mu)$. Then*

$$C_\varphi \in S_p(L^2(\mu)) \iff \varphi \in \mathcal{X}_p(\mu),$$
$$c_p\big(\mathfrak{C}_p(\varphi) + \mathfrak{D}_p(\varphi)\big) \leq \|C_\varphi\|_{S_p} \leq 4\sqrt{2}\,\mathfrak{C}_p(\varphi) + \mathfrak{D}_p(\varphi), \qquad c_p > 0.$$

Here c_p depends only on p.

Proof. Define the finite-row paraproducts $\pi_{\varphi,F}$ and the paraproduct matrix π_φ by

$$\pi_{\varphi,F} f = \sum_{I \in F} b_I \langle f, u_I\rangle h_I, \qquad [\pi_{\varphi,F}\pi^*_{\varphi,F}]_{F\times F} = G_F(\varphi),$$
$$\|\pi_{\varphi,F}\|^p_{S_p} = \operatorname{Tr}\left(G_F(\varphi)^{p/2}\right), \qquad (\pi_\varphi f)_I = b_I\langle f, u_I\rangle.$$

$$\sup_{F\in\mathcal{D}} \|\pi_{\varphi,F}\|_{S_p} < \infty \iff \pi_\varphi \in S_p, \qquad \|\pi_\varphi\|_{S_p} = \mathfrak{C}_p(\varphi). \tag{4}$$

Let D_φ be the diagonal part and ρ_I its edge difference:

$$D_\varphi h_I = d_I h_I, \qquad D_\varphi|_{\mathcal{H}_0} = 0, \qquad \rho_I = \left(|d_{I_-} - d_I|^2 + |d_{I_+} - d_I|^2\right)^{1/2},$$
$$[D_\varphi, H]h_I = (d_{I_-} - d_I)h_{I_-} - (d_{I_+} - d_I)h_{I_+},$$
$$\langle [D_\varphi, H]h_I, [D_\varphi, H]h_J\rangle = \delta_{I,J}\rho_I^2.$$

$$\|[D_\varphi, H]\|^p_{S_p} = \sum_I \rho_I^p = \mathfrak{D}_p(\varphi)^p. \tag{5}$$

For $I \subsetneq J$, $h_J(I)$ denotes the constant value of h_J on I. The multiplication matrix and the resulting form decomposition are

$$I \subsetneq J \implies h_J|_I \equiv h_J(I),$$
$$a_{I,J} := \int \varphi h_I h_J\, d\mu = \begin{cases} d_I, & I = J, \\ b_I h_J(I), & I \subsetneq J, \\ b_J h_I(J), & J \subsetneq I, \\ 0, & I \cap J = \varnothing. \end{cases} \tag{6}$$

All matrix identities here are understood on $\mathscr{F}$ before the corresponding operator extensions are established.

$$c_\varphi(f,g) = \langle [\pi_\varphi, H]f, g\rangle + \langle [(\pi_{\overline{\varphi}})^*, H]f, g\rangle + \langle [D_\varphi, H]f, g\rangle, \quad f, g \in \mathscr{F},$$
$$\pi_\varphi \in S_p,\ [D_\varphi, H] \in S_p \implies C_\varphi = [\pi_\varphi, H] + [(\pi_{\overline{\varphi}})^*, H] + [D_\varphi, H] \in S_p. \tag{7}$$

Sufficiency.

$$\varphi \in \mathcal{X}_p \Longrightarrow \|\pi_{\overline{\varphi}}\|_{S_p} = \|\pi_\varphi\|_{S_p} = \mathfrak{C}_p(\varphi),$$
$$\|C_\varphi\|_{S_p} \leq 2\,\|H\|\left(\|\pi_\varphi\|_{S_p} + \|\pi_{\overline{\varphi}}\|_{S_p}\right) + \|[D_\varphi, H]\|_{S_p}$$
$$= 4\sqrt{2}\,\mathfrak{C}_p(\varphi) + \mathfrak{D}_p(\varphi).$$

Necessity. Denote by C_{hh} the Haar compression of C, and by $\mathscr{R}$ the Schur multiplier retaining disjoint intervals. The triangular-projection bound gives

$$C = C_\varphi \in S_p, \qquad C_{hh} = P_h C|_{\mathcal{H}_h}, \qquad (\mathscr{R}T)_{I,J} = \mathbf{1}_{\{I\cap J=\varnothing\}} T_{I,J},$$
$$I = [\ell_I, r_I), \quad J = [\ell_J, r_J) \implies \mathbf{1}_{\{I\cap J=\varnothing\}} = \mathbf{1}_{\{r_I\le \ell_J\}} + \mathbf{1}_{\{r_J\le \ell_I\}},$$
$$1 < p < \infty \overset{[1,\,\S2.4]}{\Longrightarrow} \|\mathscr{R}T\|_{S_p} \le \kappa_p \|T\|_{S_p}, \qquad 0 < \kappa_p < \infty.$$

Let U be the strict ancestor part of the multiplication matrix.

$$U_{I,J} = \mathbf{1}_{\{J\subsetneq I\}} b_J h_I(J), \qquad C_{K,J} = a_{K,J_-} - a_{K,J_+} - \sigma_K a_{\widehat{K},J},$$
$$K\cap J = \varnothing \implies C_{K,J} = -\sigma_K a_{\widehat{K},J},$$
$$J \subsetneq I \implies \#\{K \in \mathrm{ch}(I) : K\cap J = \varnothing\} = 1,$$
$$\sum_{K\in\mathrm{ch}(I)} \sigma_K \mathbf{1}_{\{K\cap J=\varnothing\}} C_{K,J} = -\mathbf{1}_{\{J\subsetneq I\}} b_J h_I(J).$$

$$U = -H^*\mathscr{R}(C_{hh}), \qquad P_h \pi_\varphi|_{\mathcal{H}_h} = U^{\mathsf{t}}. \tag{8}$$

The remaining paraproduct columns are indexed by $\mathcal{H}_0$:

$$\left\|U^{\mathsf{t}}\right\|_{S_p} = \|U\|_{S_p} \le \sqrt{2}\kappa_p \|C\|_{S_p},$$
$$B = \pi_\varphi P_0, \qquad P_h C P_0 = -HB, \qquad B = -\tfrac{1}{2} H^* P_h C P_0,$$
$$\|B\|_{S_p} \le 2^{-1/2} \|C\|_{S_p}, \qquad \mathfrak{C}_p(\varphi) = \|\pi_\varphi\|_{S_p} \le (\sqrt{2}\kappa_p + 2^{-1/2}) \|C\|_{S_p}.$$

Finally, P_k projects onto the Haar functions at level k, and $\mathscr{E}_1$ extracts one-generation matrix entries:

$$P_k f = \sum_{I\in\mathcal{D}_k} \langle f, h_I\rangle h_I, \qquad V_t h_I = e^{ikt} h_I \quad (I \in \mathcal{D}_k),$$
$$\mathscr{E}_1(T) = \sum_k P_{k+1} T P_k = \frac{1}{2\pi}\int_0^{2\pi} e^{-it} V_t T V_t^*\, dt,$$
$$\|\mathscr{E}_1(T)\|_{S_p} \le \frac{1}{2\pi}\int_0^{2\pi} \|V_t T V_t^*\|_{S_p}\, dt = \|T\|_{S_p},$$
$$C_{I_-,I} = d_{I_-} - d_I, \qquad C_{I_+,I} = -(d_{I_+} - d_I), \qquad \mathscr{E}_1(C_{hh}) = [D_\varphi, H],$$
$$\mathfrak{D}_p(\varphi) \le \|C\|_{S_p}, \qquad \|\varphi\|_{\mathcal{X}_p} \le (\sqrt{2}\kappa_p + 2^{-1/2} + 1) \|C\|_{S_p}.$$

□

Proposition 2 ($p = 2$)**.** *Under the standing measure assumptions,*

$$\|C_\varphi\|_{S_2}^2 \asymp \sum_{I\in\mathcal{D}} \frac{|b_I|^2}{\mu(I)} + \sum_{I\in\mathcal{D}} \left(|d_{I_-} - d_I|^2 + |d_{I_+} - d_I|^2\right).$$

Proof.

$$\mathrm{Tr}\, G_F = \sum_{I\in F} \frac{|b_I|^2}{\mu(I)} \implies \mathfrak{C}_2(\varphi)^2 = \sum_{I\in\mathcal{D}} \frac{|b_I|^2}{\mu(I)},$$
$$(5), \quad (1) \implies \|C_\varphi\|_{S_2}^2 \asymp \mathfrak{C}_2(\varphi)^2 + \mathfrak{D}_2(\varphi)^2.$$

□

3. Finite-matrix characterization for $0 < p < \infty$

Let $\gamma_{u,v}$ denote the entries of the commutator form in the complete basis $\{e_u\}_{u\in\mathcal{I}}$. They are given by

$$\gamma_{I,J} = a_{I,J_-} - a_{I,J_+} - \sigma_I a_{\widehat{I},J}, \qquad I, J \in \mathcal{D}. \tag{9}$$

$$E, E' \in \mathscr{E}, \quad I, J \in \mathcal{D} \implies$$
$$\gamma_{I,E} = -\frac{\sigma_I b_{\widehat{I}}}{\sqrt{\mu(E)}}\mathbf{1}_{\{I\subset E\}}, \qquad \gamma_{E,J} = \frac{b_{J_-} - b_{J_+}}{\sqrt{\mu(E)}}\mathbf{1}_{\{J\subset E\}}, \qquad \gamma_{E,E'} = 0,$$
$$\gamma_{u,v} = c_\varphi(e_v, e_u), \qquad \Gamma_F = [\gamma_{u,v}]_{u,v\in F}, \qquad F \Subset \mathcal{I}.$$

Define the homogeneous space $\mathcal{Y}_p(\mu)$ from these finite sections:

$$\|\varphi\|_{\mathcal{Y}_p} = \sup_{F\Subset\mathcal{I}} \left[\operatorname{Tr}\left((\Gamma_F^*\Gamma_F)^{p/2}\right)\right]^{1/p}, \tag{10}$$
$$\mathcal{Y}_p(\mu) = \{\varphi \in L^1_{\mathrm{loc}}(\mu) : \|\varphi\|_{\mathcal{Y}_p} < \infty\}.$$

This construction retains the full commutator matrix. For $1 < p < \infty$, Theorem 1 separates its condition into $\mathfrak{C}_p$ and $\mathfrak{D}_p$.

Theorem 3. *For $\mu \in \mathfrak{M}$ and $\varphi \in L^1_{\mathrm{loc}}(\mu)$,*

$$0 < p < \infty: \qquad C_\varphi \in S_p(L^2(\mu)) \iff \varphi \in \mathcal{Y}_p(\mu), \qquad \|C_\varphi\|_{S_p} = \|\varphi\|_{\mathcal{Y}_p},$$
$$1 < p < \infty: \qquad \mathcal{Y}_p(\mu) = \mathcal{X}_p(\mu), \qquad \|\varphi\|_{\mathcal{Y}_p} \asymp_p \|\varphi\|_{\mathcal{X}_p}.$$

Proof. The finite-section bound first determines a bounded extension; the Schatten Fatou property then gives its class membership:

$$M := \sup_{F\Subset\mathcal{I}} \|\Gamma_F\|_{S_p} < \infty, \qquad \|\Gamma_F\|_{\ell^2(F)\to\ell^2(F)} \le \|\Gamma_F\|_{S_p},$$
$$f, g \in \mathscr{F}, \quad f = P_F f, \quad g = P_F g \implies |c_\varphi(f,g)| \le M\|f\|_2\|g\|_2,$$
$$\overline{\mathscr{F}} = L^2(\mu) \implies \exists!\, C \in \mathcal{B}(L^2(\mu)) : \langle Cf, g\rangle = c_\varphi(f,g), \quad \|C\| \le M,$$
$$P_F C P_F = \Gamma_F, \qquad \sup_{F\Subset\mathcal{I}} \|P_F C P_F\|_{S_p} = M \overset{\text{Fatou}}{\Longrightarrow} C \in S_p, \quad \|C\|_{S_p} \le M.$$

$$C = C_\varphi \in S_p, \quad F_n \Subset \mathcal{I}, \quad F_n \uparrow \mathcal{I} \implies \|P_{F_n} C P_{F_n} - C\|_{S_p} \longrightarrow 0,$$
$$\|C\|_{S_p} = \lim_n \|\Gamma_{F_n}\|_{S_p} \le \sup_{F\Subset\mathcal{I}} \|\Gamma_F\|_{S_p} \le \|C\|_{S_p},$$
$$1 < p < \infty, \quad (1) \implies \mathcal{Y}_p = \mathcal{X}_p, \quad \|\varphi\|_{\mathcal{Y}_p} \asymp_p \|\varphi\|_{\mathcal{X}_p}.$$

□

4. An endpoint counterexample for $0 < p \le 1$

The following measure and rank-one multiplier yield a symbol in $\mathcal{Y}_p(\mu)\backslash\mathcal{X}_p(\mu)$ for every $0 < p \le 1$. Here dx is Lebesgue measure, δ_0 is the unit point mass at 0, and $|J|$ denotes the Lebesgue length

of J.

$$I_n = [0, 2^{-n}), \quad J_n = I_n \setminus I_{n+1}, \quad n \in \mathbb{N}_0,$$

$$N_k = 2^k, \qquad n_0 = 0, \quad n_k = \sum_{r=1}^{k} 2^r = 2^{k+1} - 2, \qquad B_k = \{n_{k-1}, \dots, n_k - 1\},$$

$$\mathbb{N}_0 = \bigsqcup_{k \geq 1} B_k, \quad \#B_k = N_k, \quad w_k = \frac{3}{\pi^2 k^2}, \quad a_n = \frac{w_k}{N_k} \quad (n \in B_k),$$

$$s_0 = \tfrac{1}{2}, \qquad s_{n+1} = s_n + a_n, \qquad \sum_n a_n = \sum_k w_k = \tfrac{1}{2}, \qquad s_n \uparrow 1,$$

$$d\mu = \delta_0 + \mathbf{1}_{\mathbb{R} \setminus I_0}\, dx + \sum_{n \geq 0} \frac{s_n^{-1} - s_{n+1}^{-1}}{|J_n|} \mathbf{1}_{J_n}\, dx,$$

$$\mu \in \mathfrak{M}_\infty, \qquad \mu(\{0\}) = 1, \quad \mu(I_n) = s_n^{-1}, \qquad \varphi = \mathbf{1}_{\{0\}},$$

$$M_\varphi f = f(0)\mathbf{1}_{\{0\}}, \qquad \operatorname{rank} M_\varphi = 1 \implies \operatorname{rank} C_\varphi \leq 2 \implies \varphi \in \bigcap_{p>0} \mathcal{Y}_p(\mu).$$

The symbol Q_k below denotes the orthogonal projection onto the Haar indices in the block B_k; L_N is a strict lower-triangular matrix.

$$b_{I_n} = h_{I_n}(0) = \sqrt{\mu(I_{n+1})^{-1} - \mu(I_n)^{-1}} = \sqrt{a_n},$$

$$\langle \pi_\varphi h_{I_j}, h_{I_i} \rangle = \begin{cases} \sqrt{a_i a_j}, & i > j, \\ 0, & i \leq j, \end{cases} \qquad Q_k f = \sum_{n \in B_k} \langle f, h_{I_n} \rangle h_{I_n},$$

$$Q_k \pi_\varphi Q_k = \frac{w_k}{N_k} L_{N_k}, \qquad (L_N)_{i,j} = \mathbf{1}_{\{i>j\}},$$

$$s_j(L_N) = \left[2 \sin\left(\frac{(2j-1)\pi}{4N-2} \right) \right]^{-1}, \quad 1 \leq j \leq N-1,$$

$$\|L_N\|_{S_1} \geq \frac{2N-1}{\pi} \sum_{j=1}^{N-1} \frac{1}{2j-1} \gtrsim N \log N.$$

Block pinching in trace norm gives

$$\pi_\varphi \in S_1 \implies \|\pi_\varphi\|_{S_1} \geq \sum_{k=1}^{m} \|Q_k \pi_\varphi Q_k\|_{S_1}$$

$$\gtrsim \sum_{k=1}^{m} w_k \log N_k \asymp \sum_{k=1}^{m} \frac{1}{k} \xrightarrow[m \to \infty]{} \infty \implies \perp.$$

$$\pi_\varphi \notin S_1, \qquad 0 < p \leq 1 \implies S_p \subset S_1 \implies \mathfrak{C}_p(\varphi) = \infty,$$

$$\varphi \in \bigcap_{0<p\leq 1} \big(\mathcal{Y}_p(\mu) \setminus \mathcal{X}_p(\mu) \big).$$

5. Boundedness on $L^2(\mu)$

The operator-norm endpoint is denoted by $p = \infty$. Define $\mathfrak{A}(\varphi)$ as the local L^2 mean-oscillation seminorm and $\mathfrak{D}_\infty(\varphi)$ as the uniform edge seminorm of the diagonal coefficients d_I:

$$\mathfrak{A}(\varphi) = \sup_{Q\in\mathcal{D}} \left(\frac{1}{\mu(Q)}\int_Q |\varphi-\varphi_Q|^2\,d\mu\right)^{1/2} = \|\varphi\|_{\mathrm{bmo}_{\mu,2}}\,, \tag{11}$$

$$\mathfrak{D}_\infty(\varphi) = \sup_{I\in\mathcal{D}} \left(|d_{I_-}-d_I|^2 + |d_{I_+}-d_I|^2\right)^{1/2}. \tag{12}$$

$$\mathcal{X}_\infty(\mu) = \{\varphi\in L^1_{\mathrm{loc}}(\mu) : \mathfrak{A}(\varphi)+\mathfrak{D}_\infty(\varphi)<\infty\},$$
$$\|\varphi\|_{\mathcal{X}_\infty} = \mathfrak{A}(\varphi)+\mathfrak{D}_\infty(\varphi).$$

Theorem 4. *Let $\mu\in\mathfrak{M}$ and $\varphi\in L^1_{\mathrm{loc}}(\mu)$. Then*

$$C_\varphi\in\mathcal{B}(L^2(\mu)) \iff \varphi\in\mathcal{X}_\infty(\mu).$$

Whenever these conditions hold, write $C = C_\varphi$. One has

$$\mathfrak{A}(\varphi) \le \sqrt{3/2}\,\|C\|\,, \tag{13}$$

$$\mathfrak{D}_\infty(\varphi) \le \|C\|\,, \tag{14}$$

$$\|C\| \le 8\sqrt{2}\,\mathfrak{A}(\varphi)+\mathfrak{D}_\infty(\varphi). \tag{15}$$

$$\mathscr{E}=\varnothing \implies \mathfrak{A}(\varphi)\le\|C\|\,,$$
$$\|C\| \asymp \|\varphi\|_{\mathrm{bmo}_{\mu,2}} + \sup_{I\in\mathcal{D}}|d_I-d_{\widehat{I}}|.$$

Proof. First, local martingale approximation gives the coefficient identity. For $Q\in\mathcal{D}$, k_Q is determined by $Q\in\mathcal{D}_{k_Q}$.

$$Q\in\mathcal{D}_{k_Q}, \qquad S_{Q,n} = \sum_{\substack{J\subseteq Q\\ 0\le\log_2(\ell(Q)/\ell(J))<n}} b_J h_J = (\mathbb{E}_{k_Q+n}\varphi-\varphi_Q)\mathbf{1}_Q,$$

$$S_{Q,n}\xrightarrow[n\to\infty]{L^1(Q,\mu)}(\varphi-\varphi_Q)\mathbf{1}_Q, \qquad \|S_{Q,n}\|_2^2 = \sum_{\substack{J\subseteq Q\\ 0\le\log_2(\ell(Q)/\ell(J))<n}} |b_J|^2,$$

$$\sum_{J\subseteq Q}|b_J|^2<\infty \iff (\varphi-\varphi_Q)\mathbf{1}_Q\in L^2(\mu),$$

$$\int_Q|\varphi-\varphi_Q|^2d\mu = \sum_{J\subseteq Q}|b_J|^2\in[0,\infty].$$

$$\mathfrak{A}(\varphi)^2 = \sup_{Q\in\mathcal{D}}\frac{1}{\mu(Q)}\sum_{J\subseteq Q}|b_J|^2. \tag{16}$$

Necessity. Assume C is bounded. Its child entries control the diagonal edge seminorm:

$$C=C_\varphi\in\mathcal{B}(L^2(\mu)), \qquad N=\|C\|\,,$$
$$C_{I_-,I}=d_{I_-}-d_I, \qquad C_{I_+,I}=-(d_{I_+}-d_I),$$
$$\rho_I^2=|C_{I_-,I}|^2+|C_{I_+,I}|^2\le\|Ch_I\|_2^2\le N^2 \implies \mathfrak{D}_\infty(\varphi)\le N.$$

Fix a dyadic interval Q in the component E. For each strict ancestor A of Q, let A_Q be the child containing Q and K_A the other child.

$$Q \subset E \in \{E_-, E_+\}, \qquad A \in \mathcal{D}, \quad A \supsetneq Q,$$
$$A_Q \in \operatorname{ch}(A), \quad Q \subseteq A_Q, \qquad K_A \in \operatorname{ch}(A) \setminus \{A_Q\}, \qquad K_A \cap Q = \varnothing,$$
$$J \subseteq Q \implies \operatorname{supp} Hh_J \subseteq J, \quad \langle \varphi H h_J, h_{K_A}\rangle = 0, \quad \int \varphi h_J H^* h_{K_A}\, d\mu = \sigma_{K_A} h_A(Q) b_J.$$

$$C_{K_A,J} = -\sigma_{K_A} h_A(Q) b_J, \qquad A \supsetneq Q, \quad J \subseteq Q. \tag{17}$$

$$|h_A(Q)|^2 = \frac{1}{\mu(A_Q)} - \frac{1}{\mu(A)}, \qquad A \neq B \implies K_A \cap K_B = \varnothing,$$
$$Q^{(0)} = Q, \qquad Q^{(n+1)} = \widehat{Q^{(n)}}, \qquad Q^{(n)} \uparrow E, \qquad \mu(Q^{(n)}) \uparrow \mu(E).$$

$$\sum_{A \supsetneq Q} |h_A(Q)|^2 = \lim_{n\to\infty} \sum_{j=1}^{n} \left(\frac{1}{\mu(Q^{(j-1)})} - \frac{1}{\mu(Q^{(j)})} \right) = \frac{1}{\mu(Q)} - \frac{1}{\mu(E)}, \qquad \frac{1}{\infty} = 0. \tag{18}$$

Write $\mathcal{D}(Q)$ for the descendants of Q, including Q, and $\mathcal{D}^{\text{out}}(Q)$ for the dyadic intervals disjoint from Q. The exterior compression has the following rank-one form:

$$\mathcal{D}(Q) = \{J \in \mathcal{D} : J \subseteq Q\}, \qquad \mathcal{D}^{\text{out}}(Q) = \{K \in \mathcal{D} : K \cap Q = \varnothing\},$$
$$v_Q = \sum_{A \supsetneq Q} \sigma_{K_A} h_A(Q) h_{K_A}, \qquad F \Subset \mathcal{D}(Q),$$
$$P_{\mathcal{D}^{\text{out}}(Q)} C P_F f = -\left(\sum_{J \in F} b_J \langle f, h_J\rangle \right) v_Q,$$
$$\left\| P_{\mathcal{D}^{\text{out}}(Q)} C P_F \right\|^2 = \|v_Q\|_2^2 \sum_{J\in F} |b_J|^2 = \left(\frac{1}{\mu(Q)} - \frac{1}{\mu(E)} \right) \sum_{J\in F} |b_J|^2 \le N^2.$$

$$F \uparrow \mathcal{D}(Q) \implies \left(\frac{1}{\mu(Q)} - \frac{1}{\mu(E)} \right) \sum_{J\subseteq Q} |b_J|^2 \le N^2. \tag{19}$$

$$\mu(E) = \infty \implies \frac{1}{\mu(Q)} \sum_{J\subseteq Q} |b_J|^2 \le N^2,$$
$$E \in \mathscr{E} \implies \langle Ce_E, h_I\rangle = -\frac{\sigma_I b_{\widehat{I}}}{\sqrt{\mu(E)}} \mathbf{1}_{\{I\subset E\}}, \quad P_0 C e_E = 0.$$

For a component of finite measure, its constant basis vector supplies the remaining Carleson contribution:

$$E \in \mathscr{E} \implies \|Ce_E\|_2^2 = \frac{1}{\mu(E)} \sum_{I\subset E} |b_{\widehat{I}}|^2 = \frac{2}{\mu(E)} \sum_{J\subset E} |b_J|^2 \le N^2. \tag{20}$$

$$E \in \mathscr{E}, \quad Q \subset E \implies$$
$$\frac{1}{\mu(Q)} \sum_{J\subseteq Q} |b_J|^2 \le N^2 + \frac{1}{\mu(E)} \sum_{J\subset E} |b_J|^2 \le \frac{3}{2} N^2,$$
$$(16) \implies \mathfrak{A}(\varphi) \le \sqrt{3/2}\, N, \qquad \mathscr{E} = \varnothing \implies \mathfrak{A}(\varphi) \le N.$$

Sufficiency. Let $M_{\mathcal{D}}$ be the dyadic maximal operator. For finite F and $t>0$, $\mathcal{M}_t(F)$ consists of the inclusion-maximal intervals in $\{I\in F:|f_I|>t\}$. Doob's L^2 inequality and maximal selection give

$$\varphi\in\mathcal{X}_\infty,\quad f\in L^2(\mu),\quad M_{\mathcal{D}}f(x)=\sup_{I\ni x}\frac{1}{\mu(I)}\int_I|f|\,d\mu,\qquad \|M_{\mathcal{D}}f\|_2\le 2\,\|f\|_2\,,$$
$$F\Subset\mathcal{D},\quad t>0,\qquad \mathcal{M}_t(F)=\max_{\subseteq}\{I\in F:|f_I|>t\},$$
$$Q,R\in\mathcal{M}_t(F),\quad Q\neq R\implies Q\cap R=\varnothing,\qquad \bigcup_{Q\in\mathcal{M}_t(F)}Q\subseteq\{M_{\mathcal{D}}f>t\},$$
$$\sum_{\substack{I\in F\\ |f_I|>t}}|b_I|^2\le\sum_{Q\in\mathcal{M}_t(F)}\sum_{I\subseteq Q}|b_I|^2\le\mathfrak{A}(\varphi)^2\sum_{Q\in\mathcal{M}_t(F)}\mu(Q)\le\mathfrak{A}(\varphi)^2\mu\{M_{\mathcal{D}}f>t\}.$$

$$\begin{aligned}\|\pi_{\varphi,F}f\|_2^2=\sum_{I\in F}|b_I|^2|f_I|^2&=\int_0^\infty 2t\sum_{\substack{I\in F\\ |f_I|>t}}|b_I|^2\,dt\\ &\le\mathfrak{A}(\varphi)^2\int_0^\infty 2t\,\mu\{M_{\mathcal{D}}f>t\}\,dt\\ &=\mathfrak{A}(\varphi)^2\,\|M_{\mathcal{D}}f\|_2^2\le 4\mathfrak{A}(\varphi)^2\,\|f\|_2^2\,.\end{aligned}$$

$$\sup_{F\Subset\mathcal{D}}\|\pi_{\varphi,F}\|\le 2\mathfrak{A}(\varphi)\implies\|\pi_\varphi\|=\|\pi_{\overline{\varphi}}\|\le 2\mathfrak{A}(\varphi).\tag{21}$$

The diagonal commutator and the form decomposition now give

$$\|[D_\varphi,H]f\|_2^2=\sum_{I\in\mathcal{D}}\rho_I^2|\langle f,h_I\rangle|^2\implies\|[D_\varphi,H]\|=\sup_I\rho_I=\mathfrak{D}_\infty(\varphi),$$
$$(7),\quad(21)\implies\|C_\varphi\|\le 2\,\|H\|\left(\|\pi_\varphi\|+\|\pi_{\overline{\varphi}}\|\right)+\mathfrak{D}_\infty(\varphi)\le 8\sqrt{2}\,\mathfrak{A}(\varphi)+\mathfrak{D}_\infty(\varphi),$$
$$\sup_I|d_I-d_{\widehat{I}}|\le\mathfrak{D}_\infty(\varphi)\le\sqrt{2}\sup_I|d_I-d_{\widehat{I}}|.$$

□

Remark 5 (Paraproduct norm)**.** *Testing on normalized interval indicators gives*

$$g_Q=\frac{\mathbf{1}_Q}{\sqrt{\mu(Q)}},\qquad\|g_Q\|_2=1,$$
$$\|\pi_\varphi g_Q\|_2^2\ge\frac{1}{\mu(Q)}\sum_{J\subseteq Q}|b_J|^2,$$
$$\mathfrak{A}(\varphi)\le\|\pi_\varphi\|\le 2\mathfrak{A}(\varphi)\implies$$
$$\|C_\varphi\|\asymp\|\pi_\varphi\|+\mathfrak{D}_\infty(\varphi).$$

Remark 6 (Lebesgue measure)**.** *For Lebesgue measure, the diagonal edge condition is controlled by the dyadic mean-oscillation seminorm:*

$$d\mu=dx\implies d_I=\varphi_I,\qquad d_{I_-}-d_I=\frac{b_I}{\sqrt{\mu(I)}},\quad d_{I_+}-d_I=-\frac{b_I}{\sqrt{\mu(I)}},$$
$$\mathfrak{D}_\infty(\varphi)=\sqrt{2}\sup_I\frac{|b_I|}{\sqrt{\mu(I)}}\le\sqrt{2}\mathfrak{A}(\varphi),$$
$$\|C_\varphi\|\asymp\mathfrak{A}(\varphi)=\|\varphi\|_{\mathrm{BMO}_2^{\mathcal{D}}}\,.$$

Remark 7 (The additional diagonal condition)**.** *The following atomless example has finite local mean oscillation and an unbounded diagonal edge seminorm. Each unit interval has measure one, while its two halves may have very different masses.*

$$Q_n = [n, n+1), \quad \varepsilon_n = 2^{-2n}, \quad n \geq 1,$$

$$d\mu = \mathbf{1}_{\mathbb{R}\setminus[1,\infty)}\, dx + \sum_{n\geq 1} \big(2\varepsilon_n \mathbf{1}_{(Q_n)_-} + 2(1-\varepsilon_n)\mathbf{1}_{(Q_n)_+}\big)\, dx,$$

$$\mu \in \mathfrak{M}_\infty, \qquad \mu(\{x\}) = 0 \quad (x \in \mathbb{R}), \qquad \mu([j, j+1)) = 1 \quad (j \in \mathbb{Z}),$$

$$\varphi = \sum_{n\geq 1} \varepsilon_n^{-1/2} \mathbf{1}_{(Q_n)_-}.$$

$$I \in \mathcal{D}, \quad \ell(I) < 1 \implies \varphi|_I \equiv \varphi_I \implies \int_I |\varphi - \varphi_I|^2 d\mu = 0,$$

$$I \in \mathcal{D}, \quad \ell(I) \geq 1 \implies$$

$$\frac{1}{\mu(I)} \int_I |\varphi - \varphi_I|^2 d\mu \leq \frac{1}{\mu(I)} \int_I |\varphi|^2 d\mu = \frac{\#\{n \geq 1 : Q_n \subseteq I\}}{\mu(I)} \leq 1,$$

$$\mathfrak{A}(\varphi) \leq 1, \qquad d_{Q_n} = (1-\varepsilon_n)\varepsilon_n^{-1/2}, \qquad d_{(Q_n)_+} = 0,$$

$$\mathfrak{D}_\infty(\varphi) \geq \sup_n |d_{(Q_n)_+} - d_{Q_n}| = \sup_n (1-\varepsilon_n)2^n = \infty,$$

$$\varphi \in \mathrm{bmo}_{\mu,2} \setminus \mathcal{X}_\infty(\mu), \qquad C_\varphi \notin \mathcal{B}(L^2(\mu)).$$

6. Comparison with established results

We compare the operator, measure assumptions, and symbol condition separately. The results attributed to the references below are distinguished from consequences of Theorems 1 and 4 proved in this note.

The same downward shift and its paraproducts. López-Sánchez, Martell and Parcet [2, Theorem 2.5] study the same operator $Hh_I = h_{I_-} - h_{I_+}$. With q denoting an L^q exponent, their theorem gives

$$1 < q < 2: \qquad H \in \mathcal{B}(L^q(\mu)) \iff \sup_{I\in\mathcal{D}} \frac{m(I)}{m(\widehat{I})} < \infty,$$

$$H^* \in \mathcal{B}(L^q(\mu)) \iff \sup_{I\in\mathcal{D}} \frac{m(\widehat{I})}{m(I)} < \infty.$$

The L^2 boundedness of H is automatic. Their Lemma 5.10 gives the Carleson embedding bound used here; testing on $\mathbf{1}_Q/\sqrt{\mu(Q)}$ supplies its converse:

$$\mathfrak{A}(\varphi)^2 = \sup_Q \frac{1}{\mu(Q)} \sum_{I\subseteq Q} |b_I|^2, \qquad \mathfrak{A}(\varphi) \leq \|\pi_\varphi\| \leq 2\mathfrak{A}(\varphi).$$

Thus the paraproduct boundedness ingredient is an established result. Theorem 4 concerns the additional symbol information needed for the commutator on $L^2(\mu)$:

$$\|C_\varphi\| \asymp \mathfrak{A}(\varphi) + \mathfrak{D}_\infty(\varphi).$$

Martingale BMO and the diagonal condition. Treil [3, Proposition 5.1 and Theorem 5.3] proves BMO sufficiency for commutators with bounded martingale transforms and a converse under mixing hypotheses, with the stated corrections for finite-measure tail atoms. Such transforms preserve martingale difference blocks. Here H changes the dyadic level, so that theorem does not directly identify its commutator symbols.

To compare the sufficient BMO condition with our exact condition, set

$$J(\varphi) = \sup_{I\in\mathcal{D}} |\varphi_I - \varphi_{\widehat{I}}|,$$

$$\mathfrak{B}(\varphi) = \sup_{I\in\mathcal{D}} \left(\frac{1}{\mu(I)}\int_I |\varphi - \varphi_{\widehat{I}}|^2 d\mu\right)^{1/2}.$$

The space $\{\mathfrak{B} < \infty\}$ is martingale $\mathrm{BMO}_{\mu,2}$, whereas $\{\mathfrak{A} < \infty\}$ is $\mathrm{bmo}_{\mu,2}$. Directly,

$$\frac{1}{\mu(I)}\int_I |\varphi - \varphi_{\widehat{I}}|^2 d\mu = \frac{1}{\mu(I)}\int_I |\varphi - \varphi_I|^2 d\mu + |\varphi_I - \varphi_{\widehat{I}}|^2,$$

$$\max\{\mathfrak{A}, J\} \le \mathfrak{B} \le (\mathfrak{A}^2 + J^2)^{1/2},$$

$$|d_I - \varphi_I| \le \max\{|\varphi_{I_-} - \varphi_I|, |\varphi_{I_+} - \varphi_I|\} \le J,$$

$$|d_I - d_{\widehat{I}}| \le 3J, \qquad \mathfrak{D}_\infty \le 3\sqrt{2}J,$$

$$\|\varphi\|_{\mathcal{X}_\infty} \le \mathfrak{A} + 3\sqrt{2}J \le (1 + 3\sqrt{2})\mathfrak{B},$$

$$\mathrm{BMO}_{\mu,2} \subseteq \mathcal{X}_\infty(\mu) \subseteq \mathrm{bmo}_{\mu,2}.$$

These inclusions are asserted without a strictness claim for the first inclusion. The last example of the preceding section shows that $\mathfrak{A} < \infty$ alone need not imply boundedness.

The sibling shift and the closest symbol characterization. Let I^s denote the sibling of I and define $\mathcal{S}$ by

$$\mathcal{S}h_{I_-} = -h_{I_+}, \qquad \mathcal{S}h_{I_+} = h_{I_-}, \qquad \mathcal{S}|_{\mathcal{H}_0} = 0.$$

Borges, Conde-Alonso, Pipher and Wagner [4, Theorem A] prove BMO upper bounds for this sibling shift and show that a commutator lower bound for the full BMO norm fails in general. D'Emilio, Lin, Wagner and Wick [5, Theorem D; Theorem 4.7] give its exact symbol characterization. In the atomless, sibling-balanced setting, with infinite measure on both dyadic components, the L^2 specialization is

$$[\mu]_{\mathrm{sib}} := \sup_{I\in\mathcal{D}} \frac{m(I)}{m(I^s)} < \infty, \qquad \beta_I = d_I - d_{I^s},$$

$$[M_\varphi, \mathcal{S}] \in \mathcal{B}(L^2(\mu)) \iff \mathfrak{A}(\varphi) < \infty, \quad \beta \in \ell^\infty.$$

Thus the use of $d_I = \langle \varphi, h_I^2\rangle$ already occurs in that work. The respective differences follow the respective shift actions:

$$\|[D_\varphi, H]\| = \sup_I \left(|d_{I_-} - d_I|^2 + |d_{I_+} - d_I|^2\right)^{1/2},$$

$$\|[D_\varphi, \mathcal{S}]\| = \sup_I |d_I - d_{I^s}|.$$

Reference	Shift	Measure assumptions	Boundedness result on L^2
Borges et al. [4]	$\mathcal{S}$	Sibling balanced	$\|[M_\varphi, \mathcal{S}]\| \lesssim_{\mathrm{sib}} \mathfrak{B}(\varphi)$; full BMO lower bound fails.
D'Emilio et al. [5]	$\mathcal{S}$	Atomless, sibling balanced	$\mathfrak{A}(\varphi) < \infty$ and $\sup_I |d_I - d_{I^s}| < \infty$.
Theorem 4	H	$\mu \in \mathfrak{M}$	$\mathfrak{A}(\varphi) + \mathfrak{D}_\infty(\varphi) < \infty$.

The sibling-shift rows are read with $\mu(E_-) = \mu(E_+) = \infty$. The subscript sib allows dependence on $[\mu]_{\mathrm{sib}}$. The table compares distinct shifts. It does not identify the two commutator symbol spaces for a fixed measure.

Schatten–Besov results and the scope of our criterion. For regular dyadic martingales with Lebesgue measure, the classical Chao–Peng paraproduct theorem [6], recalled in [7, Theorem 1.1], gives a Besov characterization for every $0 < p < \infty$. The norm equivalence also follows from the scalar, $d = 2$ case of [7, Theorem 1.2]:

$$\|\varphi\|_{B_p^{\mathcal{D}}} := \left(\sum_{I \in \mathcal{D}} |I|^{-p/2} |b_I|^p \right)^{1/p},$$

$$d\mu = dx, \quad 0 < p < \infty : \qquad \|\pi_\varphi\|_{S_p} = \mathfrak{C}_p(\varphi) \asymp_p \|\varphi\|_{B_p^{\mathcal{D}}}.$$

Wei and Zhang's theorem also treats operator-valued symbols on regular d-adic filtrations. Our symbols are scalar, while the measure is arbitrary. For the common scalar Lebesgue case,

$$\mathfrak{D}_p(\varphi) = \sqrt{2}\, \|\varphi\|_{B_p^{\mathcal{D}}},$$

$$1 < p < \infty : \qquad \|C_\varphi\|_{S_p} \asymp_p \|\varphi\|_{B_p^{\mathcal{D}}} \qquad \text{by Theorem 1.}$$

For arbitrary μ, $\mathfrak{C}_p$ retains the interval correlations through G_F, and Theorem 1 has the range $1 < p < \infty$. The atomic example in Section 4 prevents extending this separated criterion to $p \le 1$ for all measures; it does not conflict with the regular-filtration paraproduct theorem. Theorem 3, valid for all $p > 0$, is a finite-matrix reformulation and supplies no coefficient-only Besov characterization in that endpoint range.

References

[1] J. Parcet, M. de la Salle and E. Tablate, *The local geometry of idempotent Schur multipliers*, Forum Math. Pi **13** (2025), e14, doi:10.1017/fmp.2025.6, §2.4.

[2] L. D. López-Sánchez, J. M. Martell and J. Parcet, *Dyadic harmonic analysis beyond doubling measures*, Adv. Math. **267** (2014), 44–93, doi:10.1016/j.aim.2014.08.001.

[3] S. Treil, *Commutators, paraproducts and BMO in non-homogeneous martingale settings*, Rev. Mat. Iberoam. **29** (2013), no. 4, 1325–1372, doi:10.4171/RMI/759.

[4] T. Borges, J. M. Conde-Alonso, J. Pipher and N. A. Wagner, *Commutator estimates for Haar shifts with general measures*, J. Funct. Anal. **289** (2025), no. 5, Article 110945, doi:10.1016/j.jfa.2025.110945.

[5] F. D'Emilio, Y. Lin, N. A. Wagner and B. D. Wick, *Optimal sparse bounds and commutator characterizations without doubling*, preprint (2025), arXiv:2510.26505v1.

[6] J.-A. Chao and L. Peng, *Schatten classes and commutators on simple martingales*, Colloq. Math. **71** (1996), no. 1, doi:10.4064/cm-71-1-7-11.

[7] Z. Wei and H. Zhang, *On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators*, preprint (2024), arXiv:2407.14988v3.